ELECTRO RANGERS | AURA2026

TITLE OF THE PROJECT:

Protecting People's Lives from Electrocution Caused by Electrical Equipment on Waterlogged Streets

A smart sensing and alert system that detects live current in street water and warns both pedestrians and the power utility before an accident happens.

The Problem

During heavy monsoon rain, many city streets stay waterlogged for hours. Street lights, electrical poles, junction boxes and feeder pillars stand in or beside this water. If insulation fails or a live wire snaps and falls, electric current can leak into the water.

The danger is **invisible**. The water looks like any other puddle, there is no sound and no warning, and people only find out when they touch it or step in. Pedestrians, hand-rickshaw pullers, cyclists and two-wheeler riders are the most exposed. The electricity utility often learns about the fault only after someone has already been hurt.

> **Our idea in one line**
> Place a low-cost sensing unit on every high-risk pole. It checks the water for electric leakage, warns people on the spot with a siren and flashing light, and instantly alerts the substation so the dangerous line can be switched off.

Project Objectives

Detect

Sense dangerous voltage leaking into street water, but only when water is actually present.

Warn

Alert people nearby within seconds using a buzzer and a light signal on the pole.

Isolate

Notify the substation so that power to the faulty section can be cut quickly and safely.

Methodology

The system works in seven stages, from detecting the hazard to cutting the power. Every stage is built from simple, low-cost hardware that can be assembled and tested as a student prototype.

1

Sense

A stainless steel electrode measures the small voltage that appears in the water when current is leaking. A water-level (float) sensor confirms that the street is really waterlogged.

2

Isolate and condition

An optocoupler and op-amp circuit separates the sensor from the microcontroller and scales the tiny signal into a safe range. The electronics stay protected even if the water is live.

3

Decide

An ESP32 microcontroller raises the alarm only when two checks pass: water is present AND voltage is present.

4

Warn locally

A buzzer, a flashing LED and a STOP signal (toll-gate style barrier) switch on immediately, so people nearby are warned without waiting for any remote action.

5

Travellers stopped

Seeing the STOP signal, travellers will not pass through the electrified street, which keeps them out of danger.

6

Communicate

A GSM or LoRa module sends the pole ID, voltage reading, water level and time to the substation. A regular "heartbeat" message proves the unit is still working.

7

Respond

The substation dashboard shows the alarming pole on a map. After operator confirmation (or an automatic trip at a high threshold), a relay opens the breaker and cuts power to that feeder section, covering the whole affected area. A repair crew is sent to the location.

Safety and Reliability Features

Two-condition check

Water AND voltage must both be present before any alarm or power cut.

Isolated sensing

The optocoupler keeps the microcontroller electrically separate from the water.

Power Supply

From the power supply and a battery keeps the unit working during power cuts and storms.

Fault self-check

If a pole stops sending its heartbeat, the substation flags a sensor fault.

Proposed Diagram of the Model

The diagram below shows the complete model: the sensing unit mounted on the pole, the wireless alert, and the substation response, followed by the seven working steps.

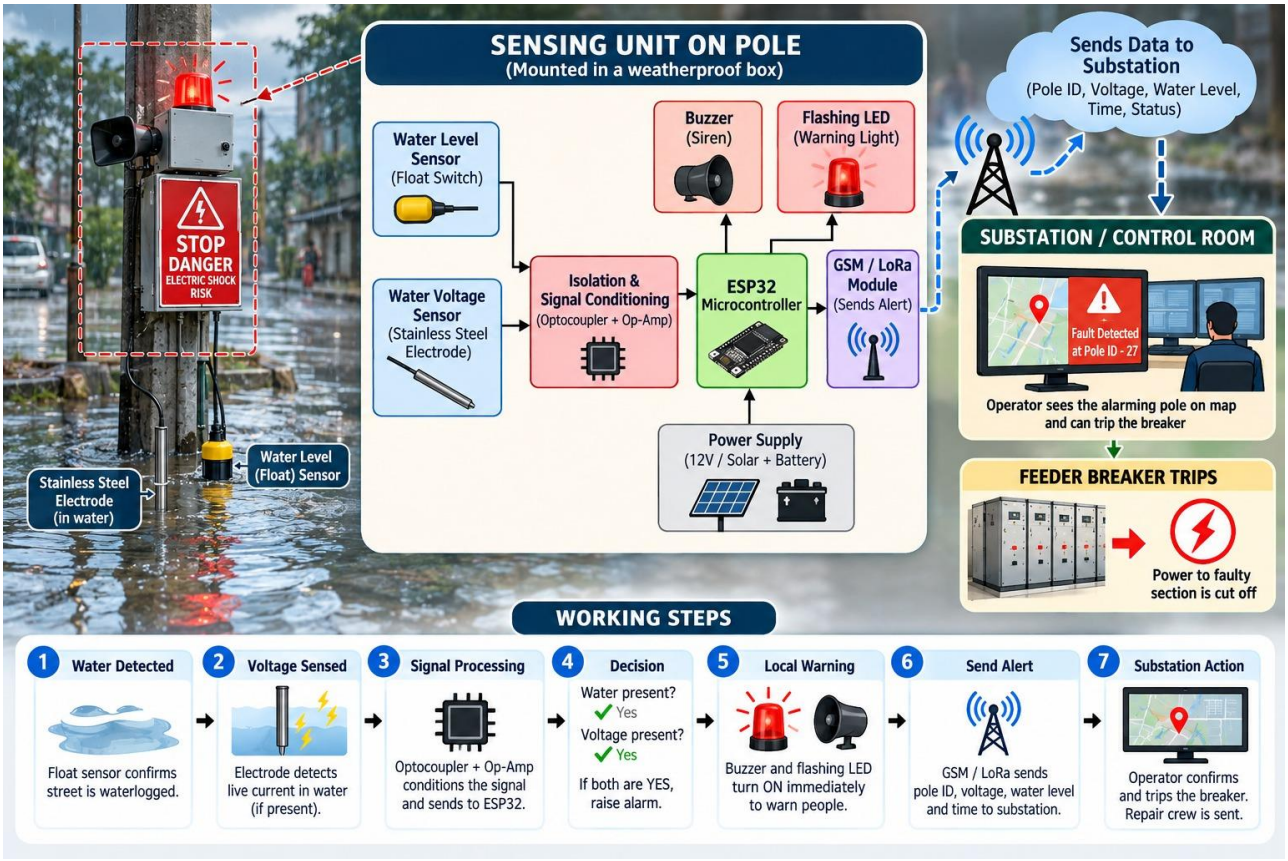


Figure : Proposed model of the waterlogged-street electrocution protection system

Proposed Outcome of the Project

Lives saved
On-the-spot warning and a quick power cut greatly reduce the chance of electric shock in flooded streets.

Live wire detection
Broken or fallen live wires and leaking equipment are detected automatically and located by pole ID.

Faster response
The utility gets alerts in seconds instead of waiting for public complaints, so repair crews reach faults sooner.

Safe route information
Alarm status helps people and traffic authorities avoid dangerous streets, reducing delays for bikes, hand rickshaws, scooters and pedestrians.

Low cost and scalable
Built from simple, low-cost parts, so every high-risk pole in a city can be covered.

Data for planning
Alarm records show which areas are most flood-prone and fault-prone, helping maintenance planning.

Compatibility
Built for compatible with pole carrying overhead cable and Distribution box.

Conclusion. By combining simple sensors, a microcontroller and a wireless link, this project turns an invisible street hazard into an instant warning and a fast power cut, helping to save lives every monsoon at a very low cost per pole.

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